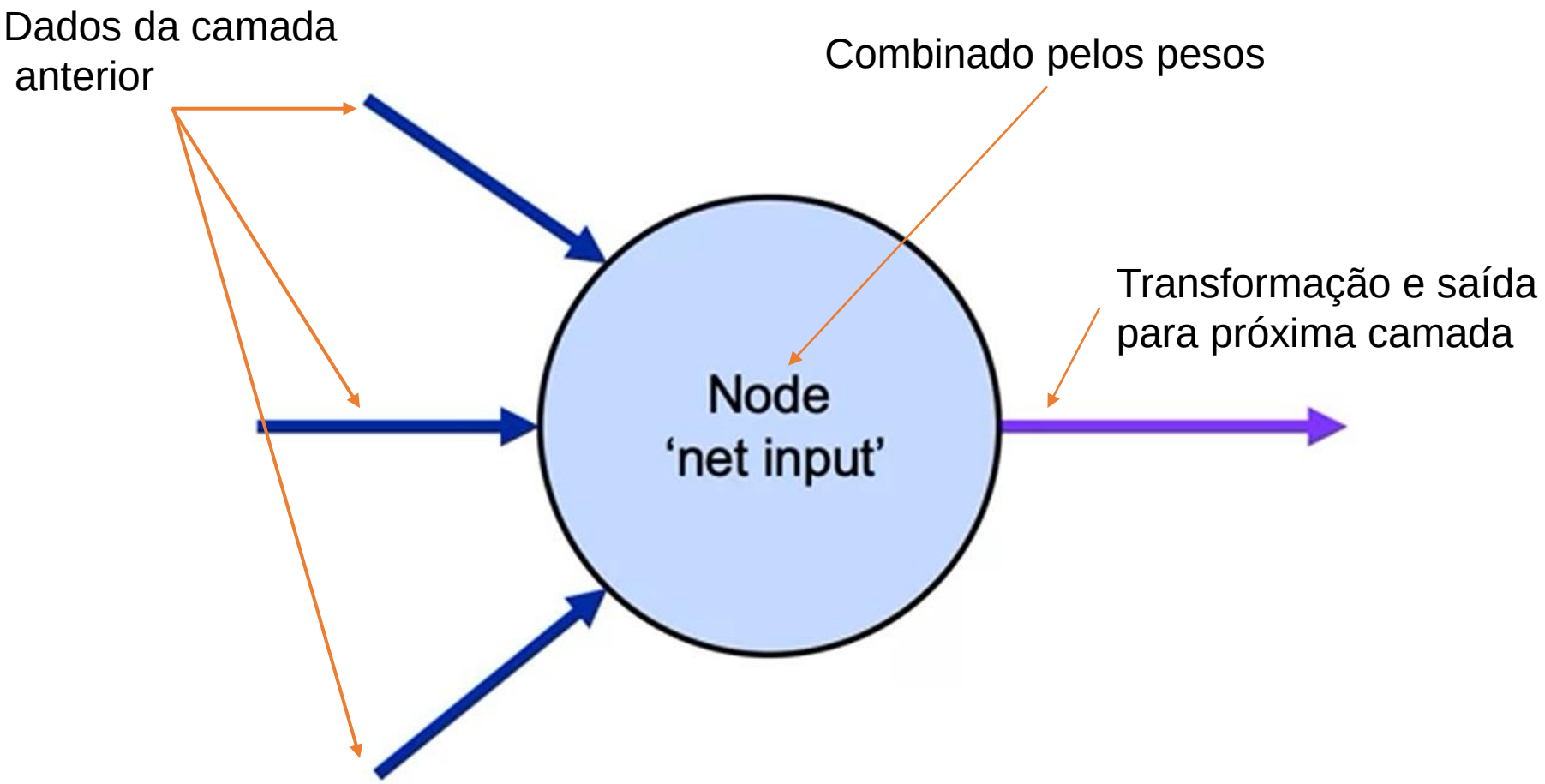
Visualização Básica do Neurônio



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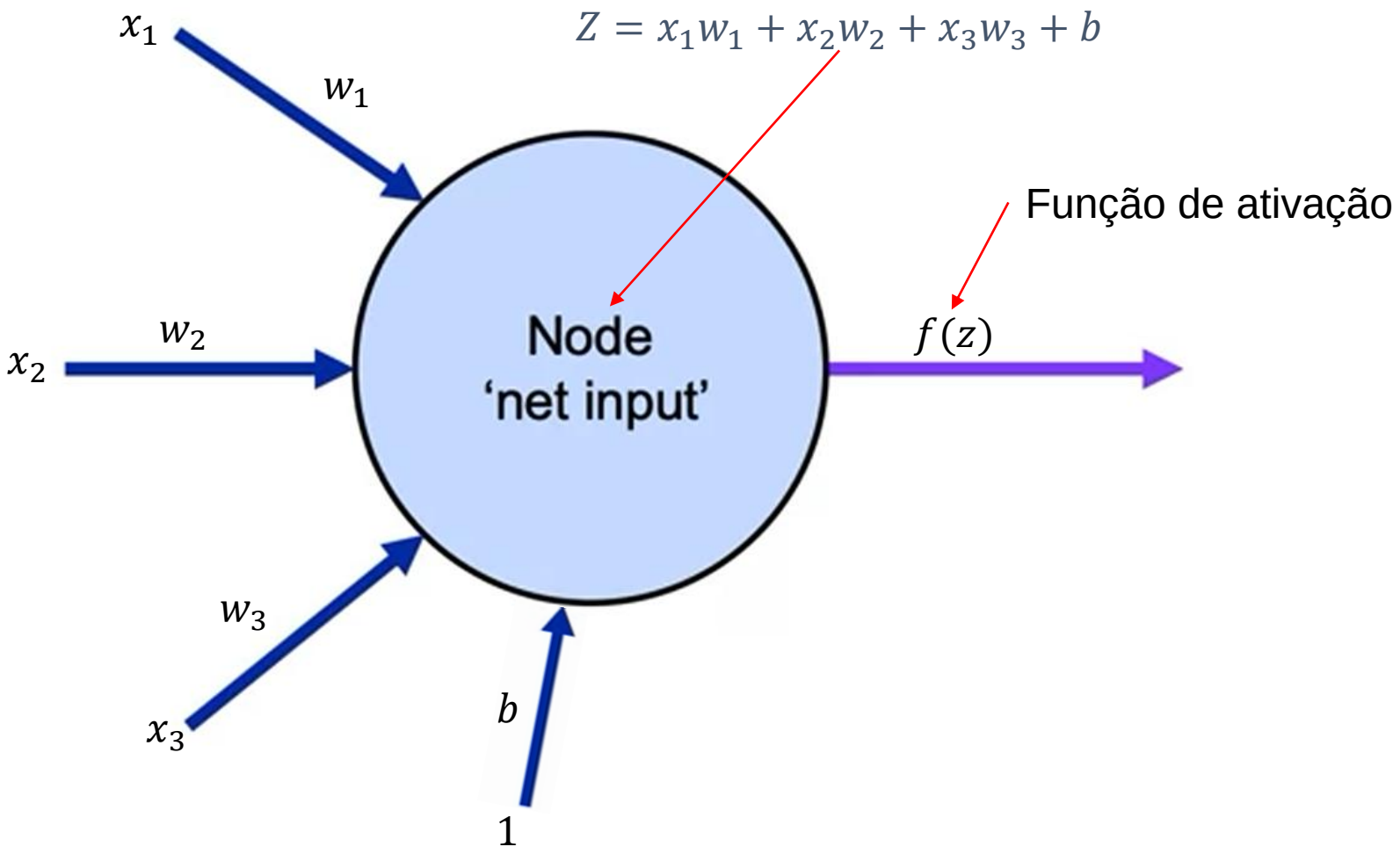


Em Notação Vetorial

- z = entrada da rede (“net input”)
- b = viés (“bias term”)
- f = função de ativação
- a = saída para a próxima camada

Relação com a Regressão Logística

Quando escolhemos $f(z) = \frac{1}{1+e^{-z}}$

$$z = b + \sum_{i=1}^{m} x_i w_i = x_1 w_1 + x_2 w_2 + \cdots + x_m w_m + b$$

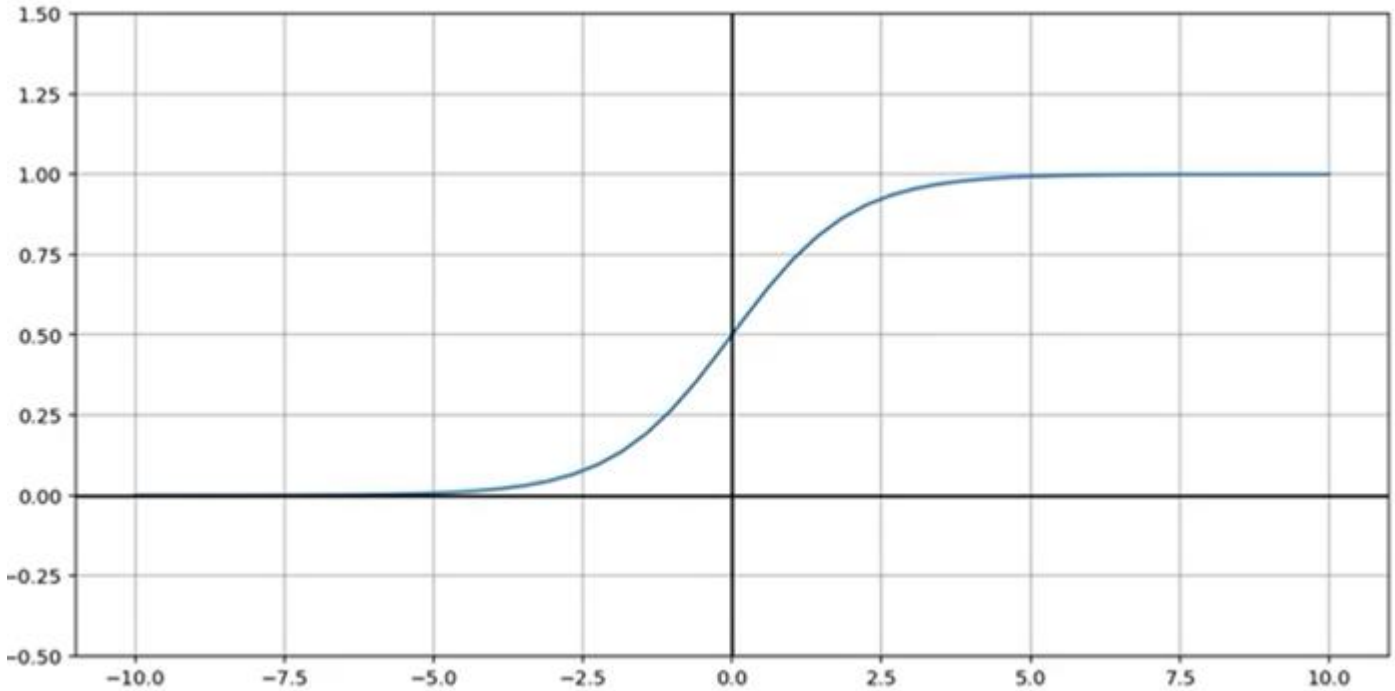
Então o neurônio é simplesmente uma unidade da regressão logística.

- Pesos $\leftrightarrow$ coeficientes
- Entradas $\leftrightarrow$ variáveis
- Bias $\leftrightarrow$ constante

Relação com a Regressão Logística

Esta é chamada função sigmoide

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



Propriedade interessante da função sigmoide

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\frac{d}{dx} \cdot \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

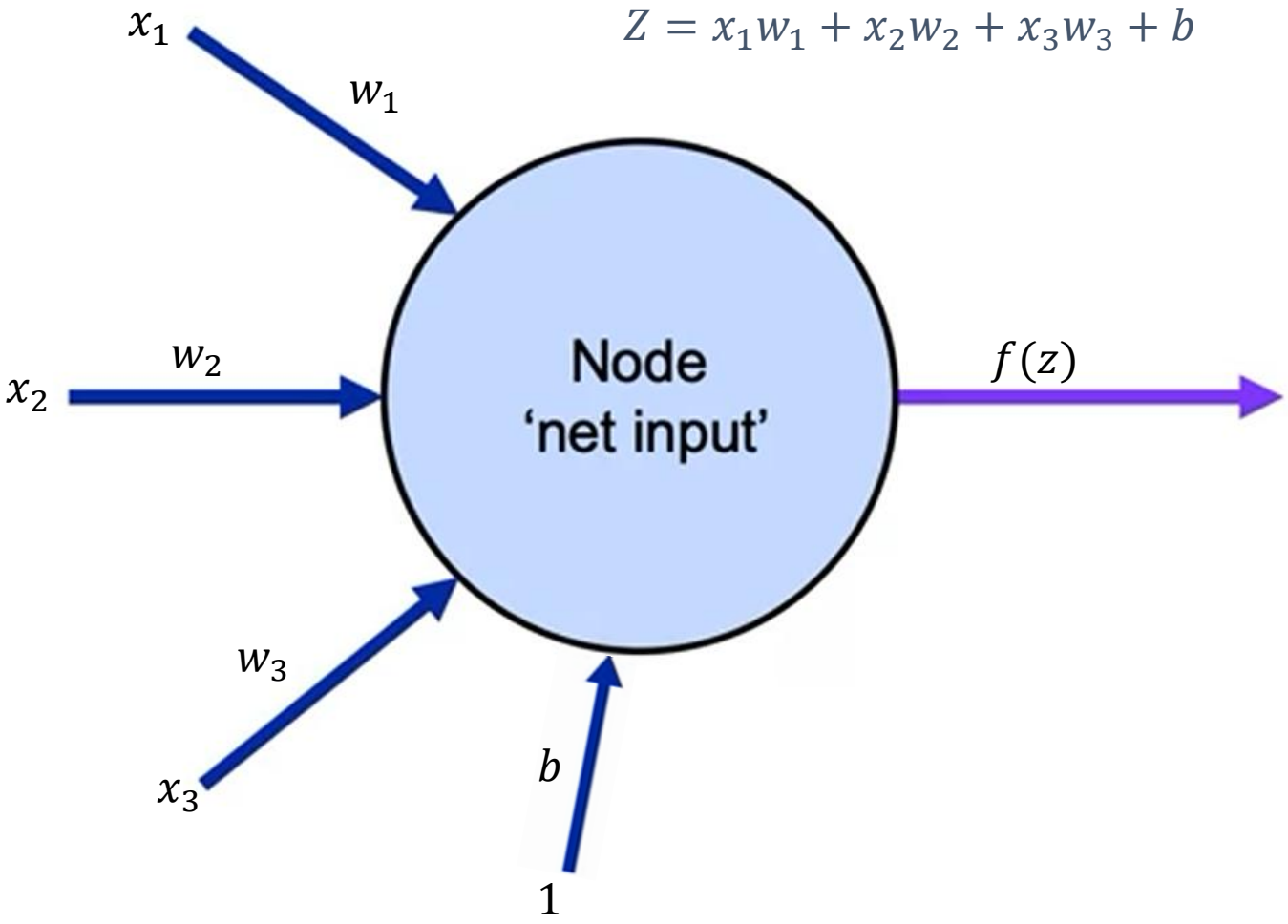
$$\sigma'(z) = \frac{0-(-e^{-z})}{(1+e^{-z})^2} = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1+e^{-z}-1}{(1+e^{-z})^2} = \frac{\cancel{1+e^{-z}}}{(1+e^{-z})^{\cancel{2}}} - \frac{1}{(1+e^{-z})^2}$$

$$= \frac{1}{1+e^{-z}} - \frac{1}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}}\left(1 - \frac{1}{1+e^{-z}}\right)$$

$$\sigma'(z) = \sigma(z)(1-\sigma(z))$$

Exemplo de cálculo

$$Z = x_1 w_1 + x_2 w_2 + x_3 w_3 + b$$



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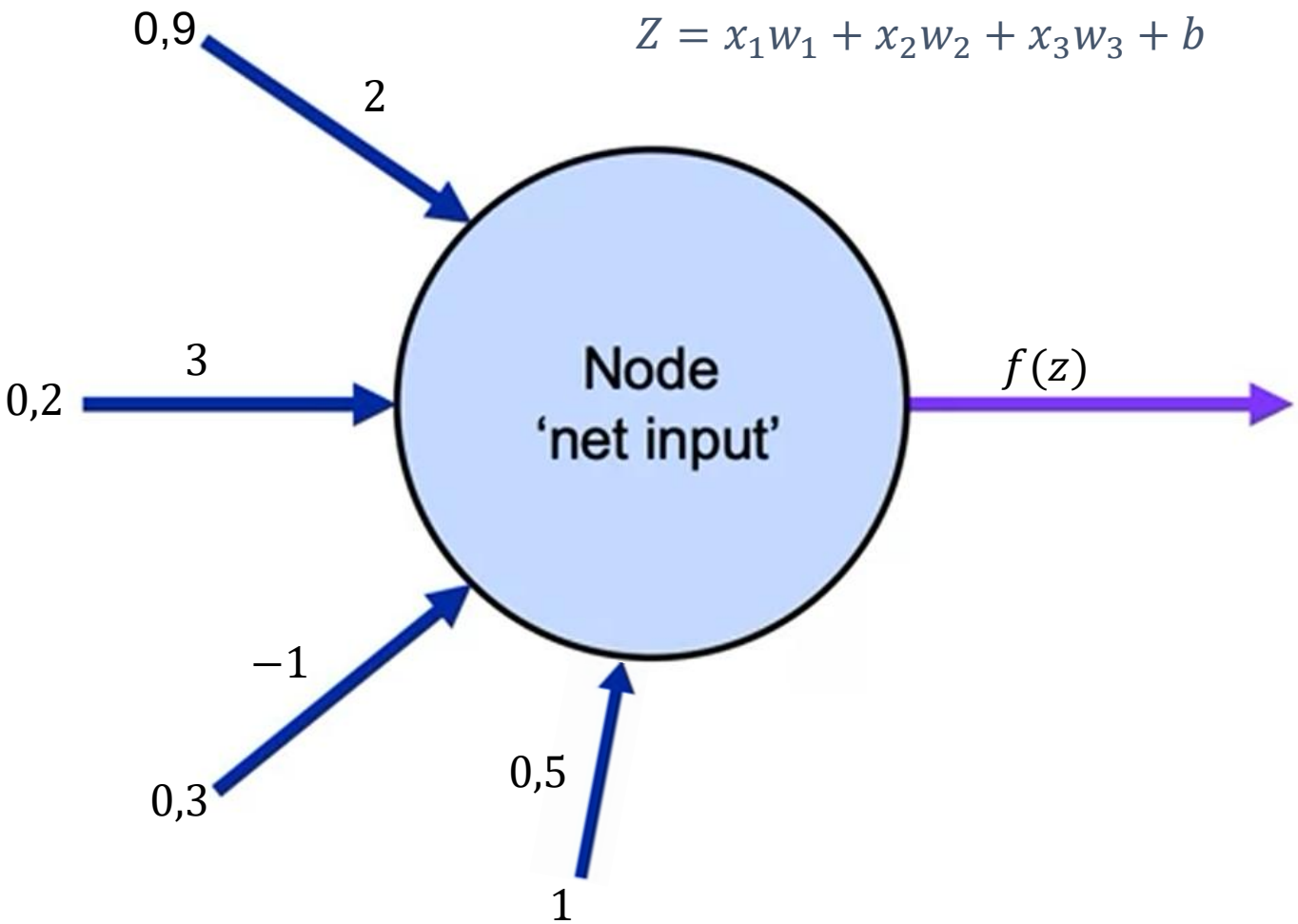
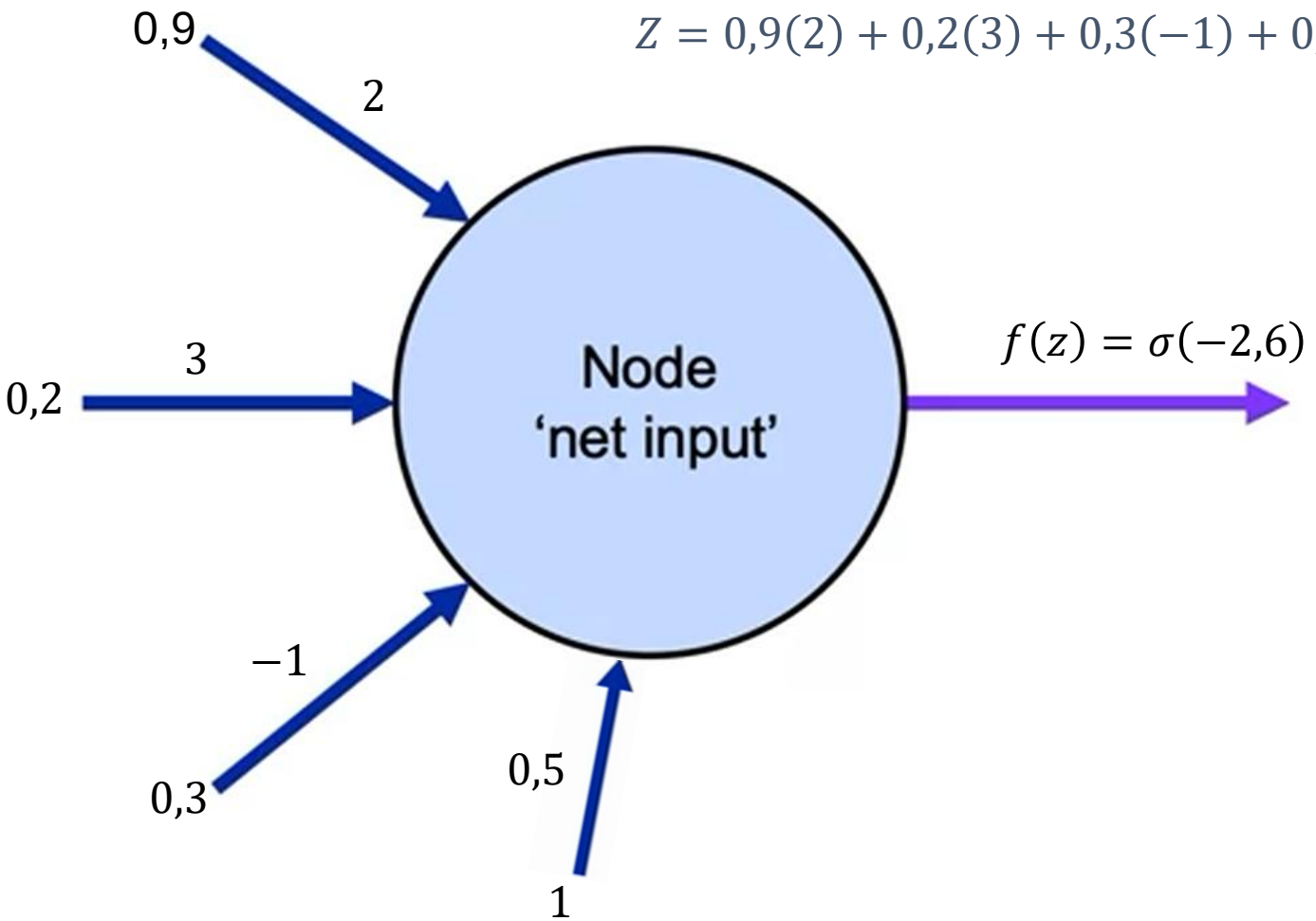


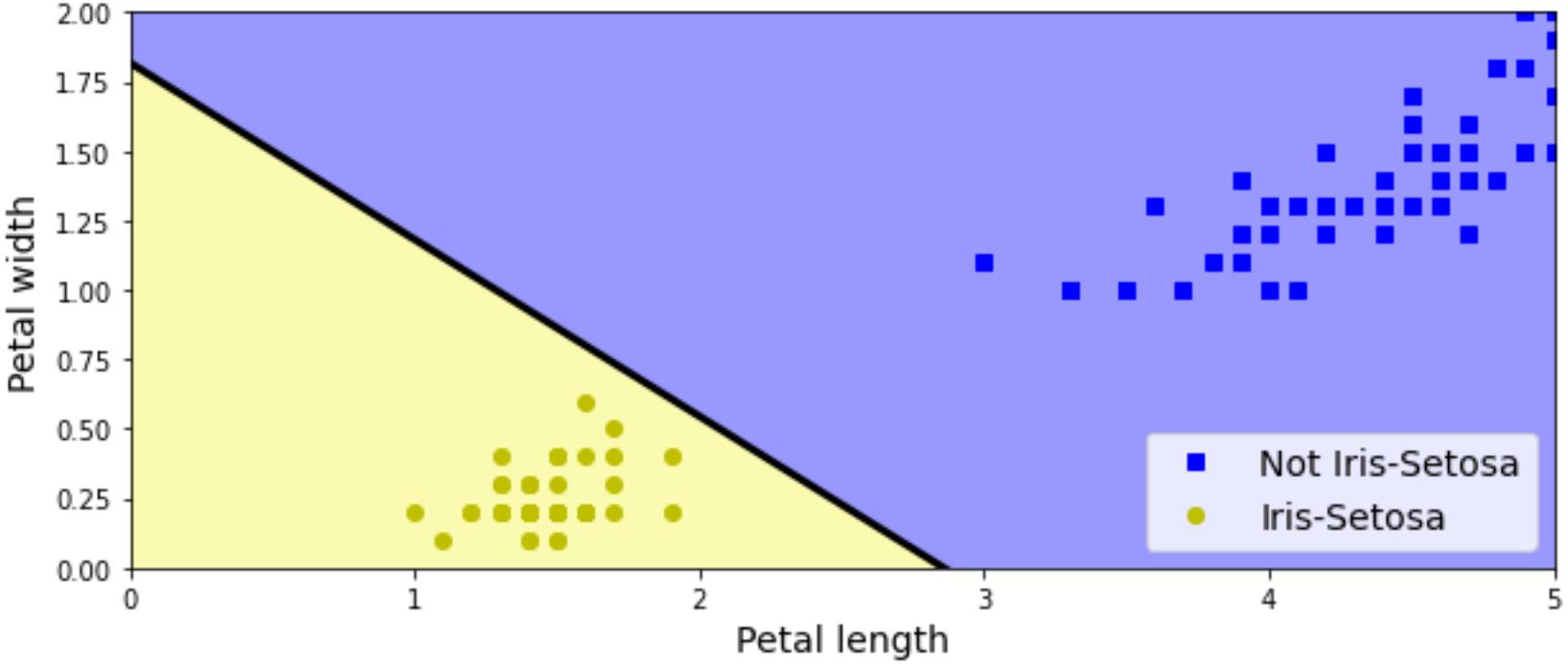
Exemplo de cálculo

$$Z = 0{,}9(2) + 0{,}2(3) + 0{,}3(-1) + 0{,}5 = 2{,}6$$

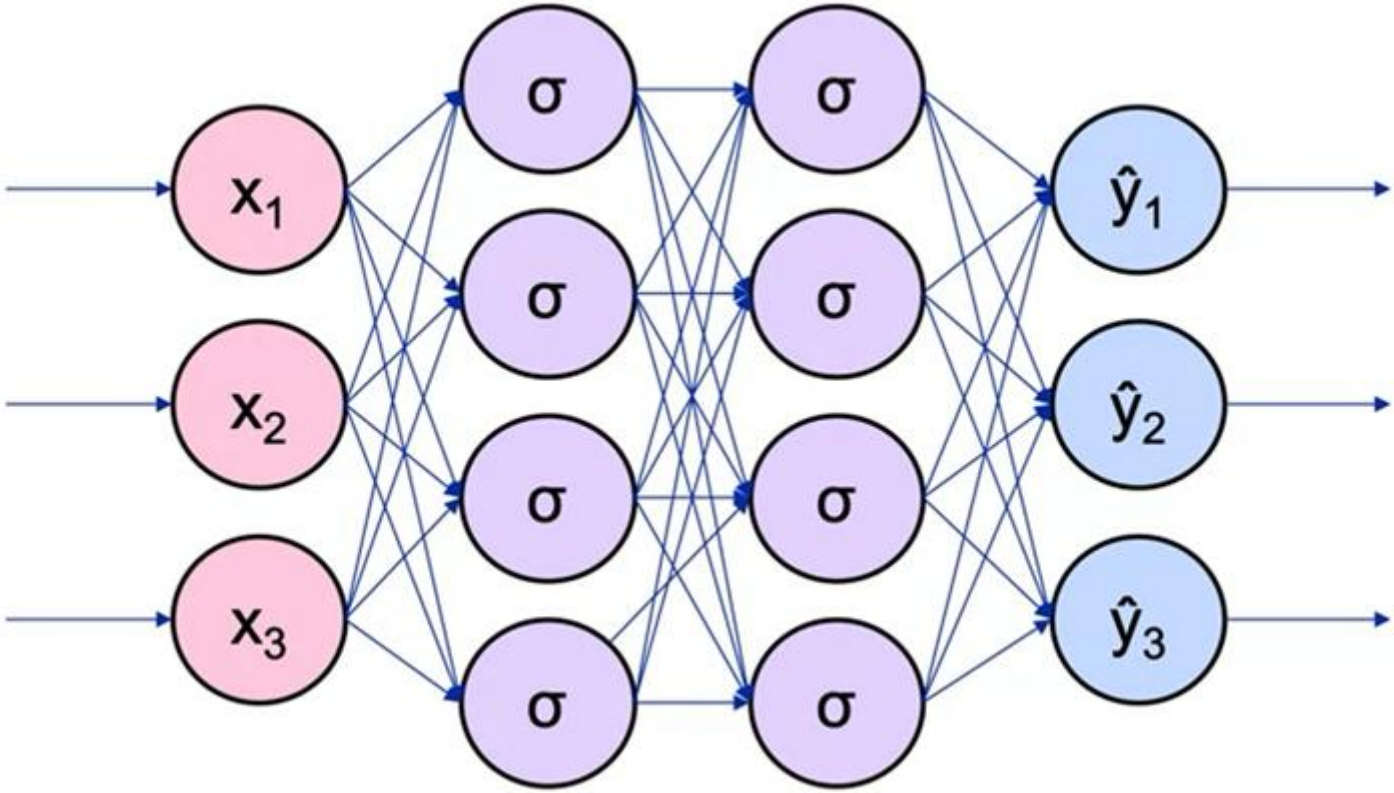


$$f(z) = \sigma(-2{,}6) = 0{,}93$$

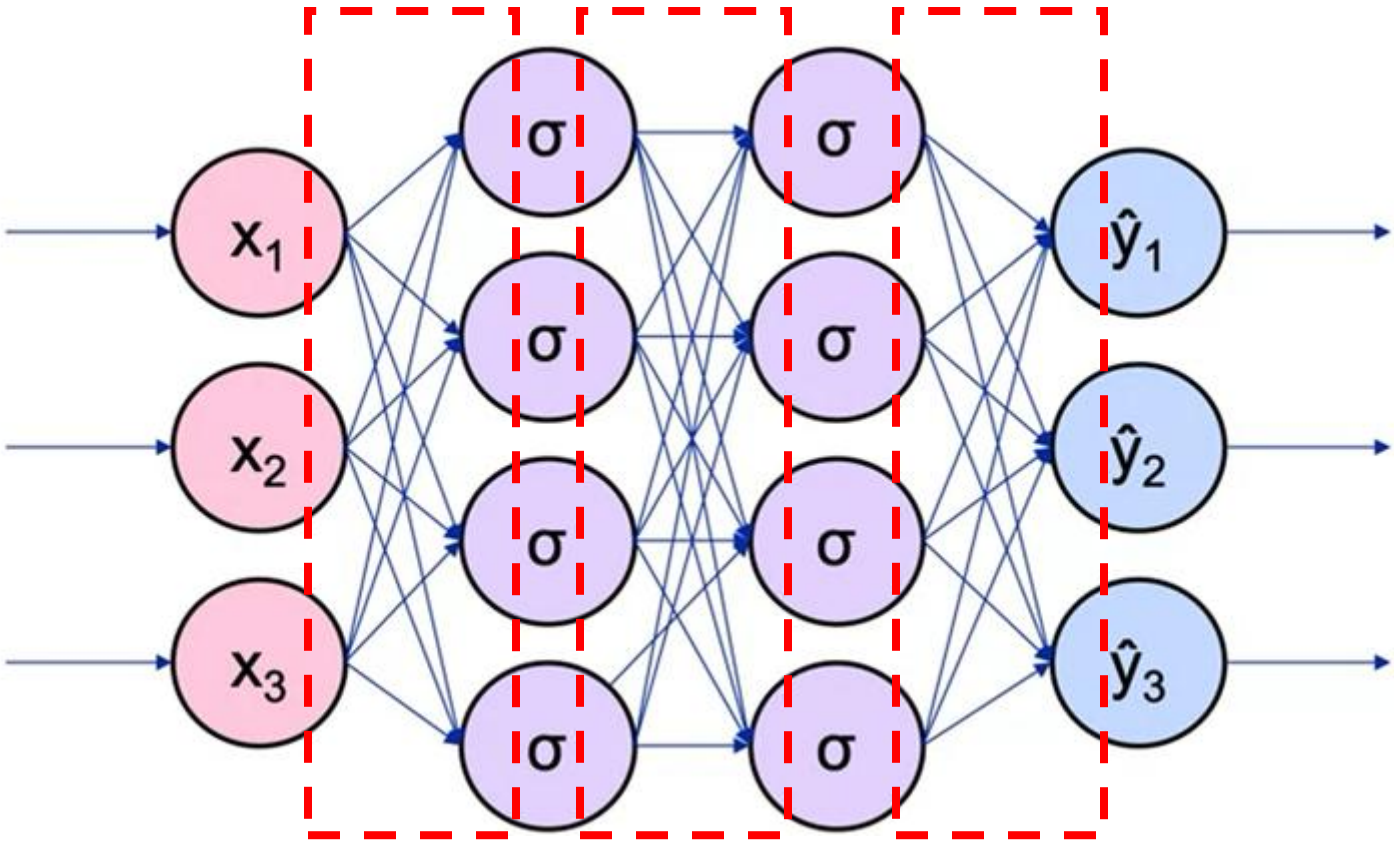
Exemplo de aplicação



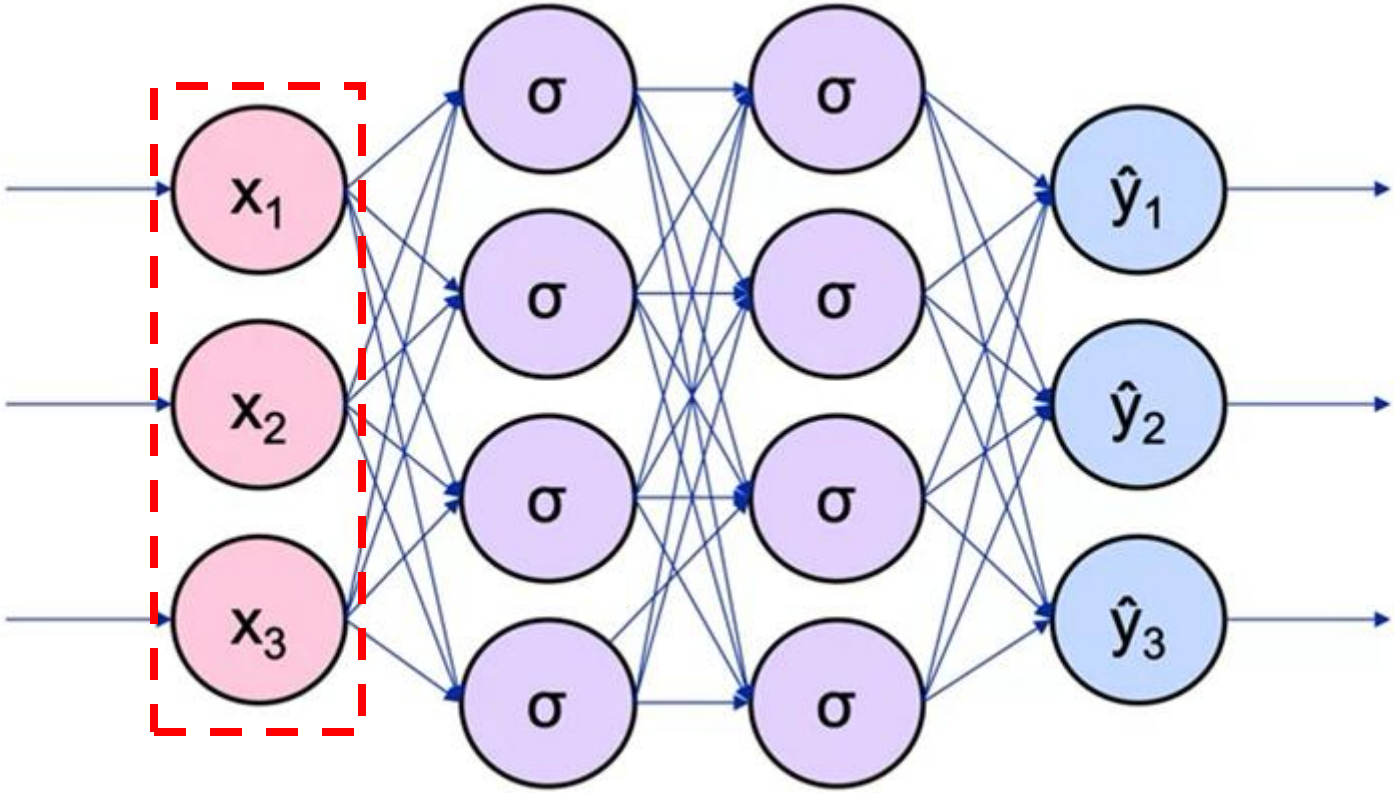
Rede Feedforward: Multilayer Perceptron



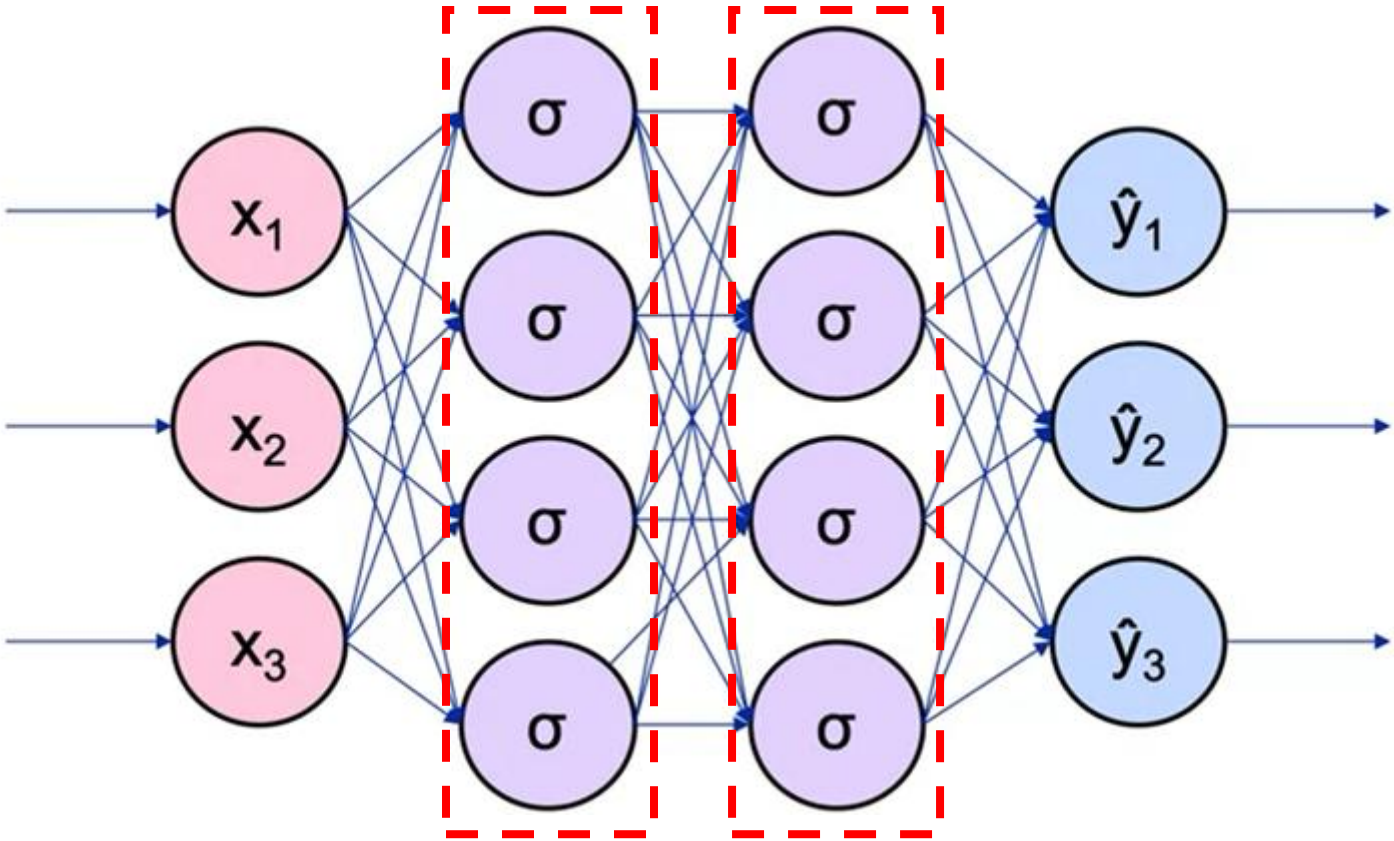
Multilayer Perceptron: Pesos



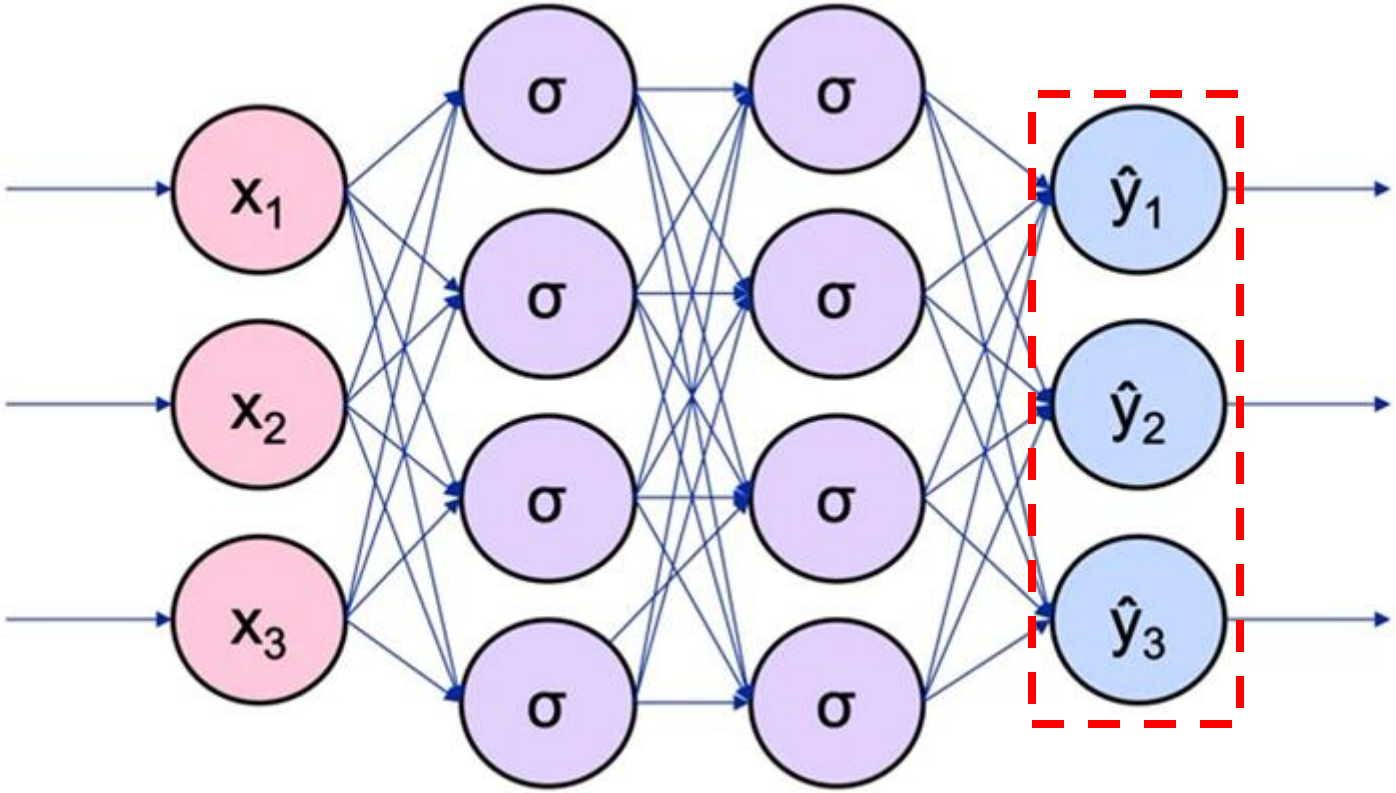
Camada de Entrada



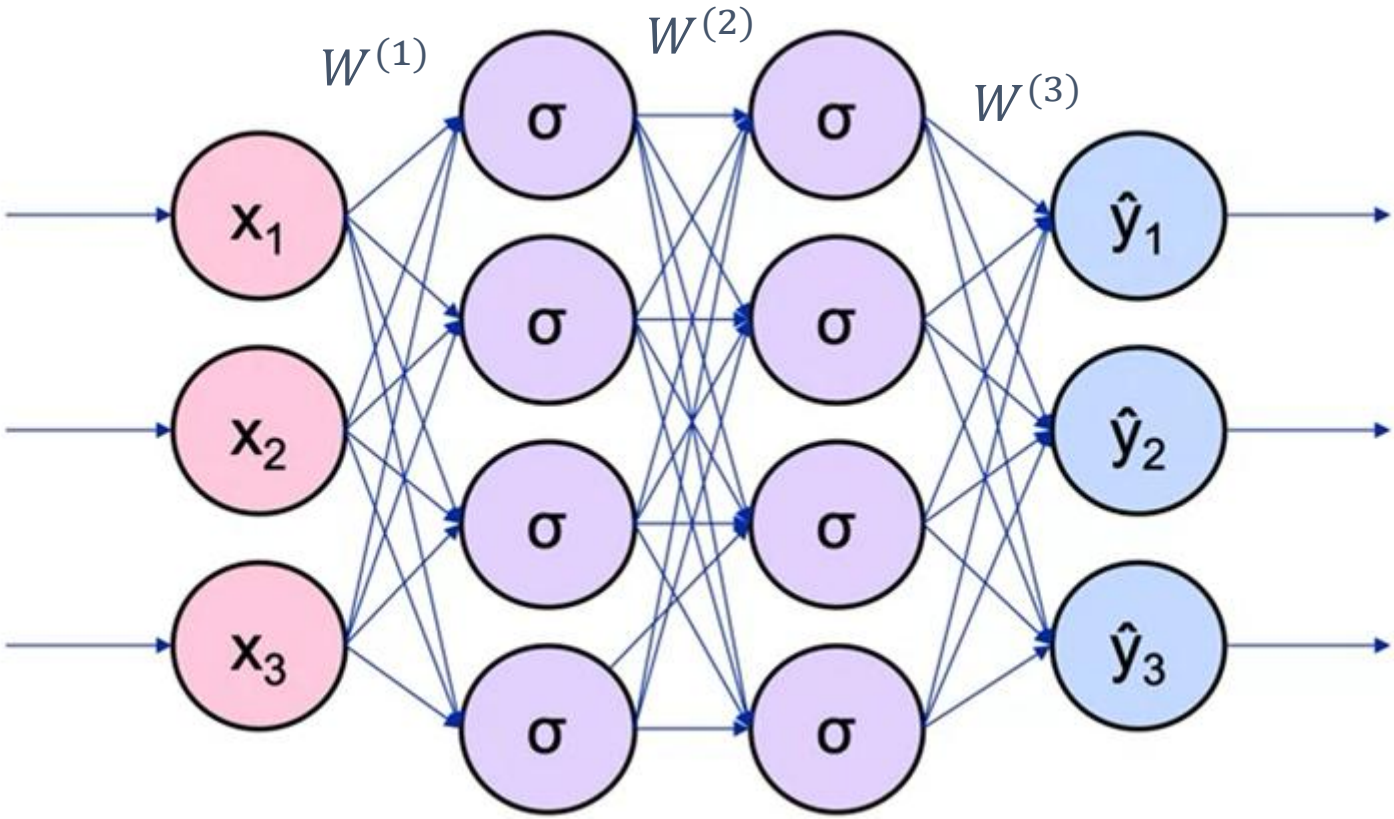
Camadas Ocultas



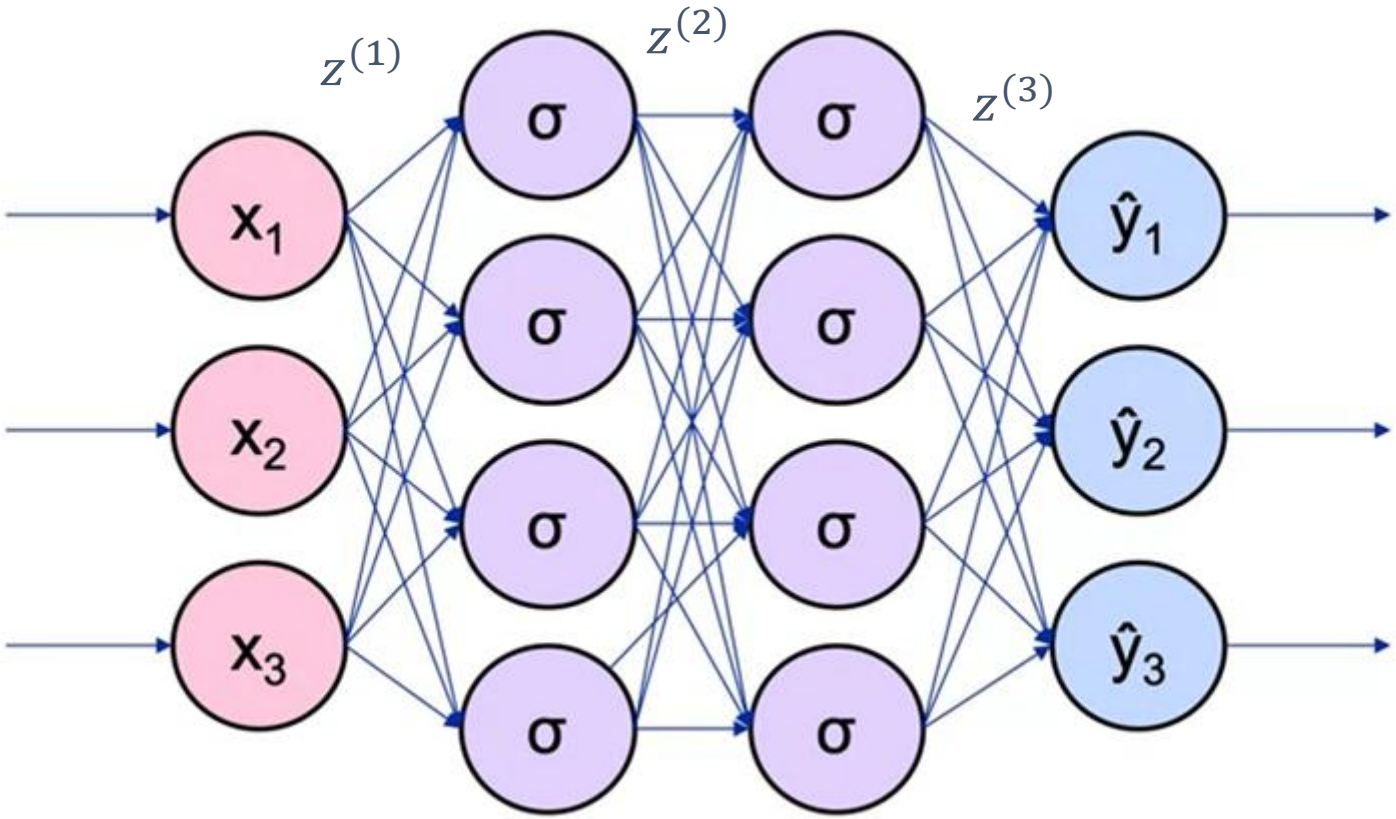
Camada de Saída



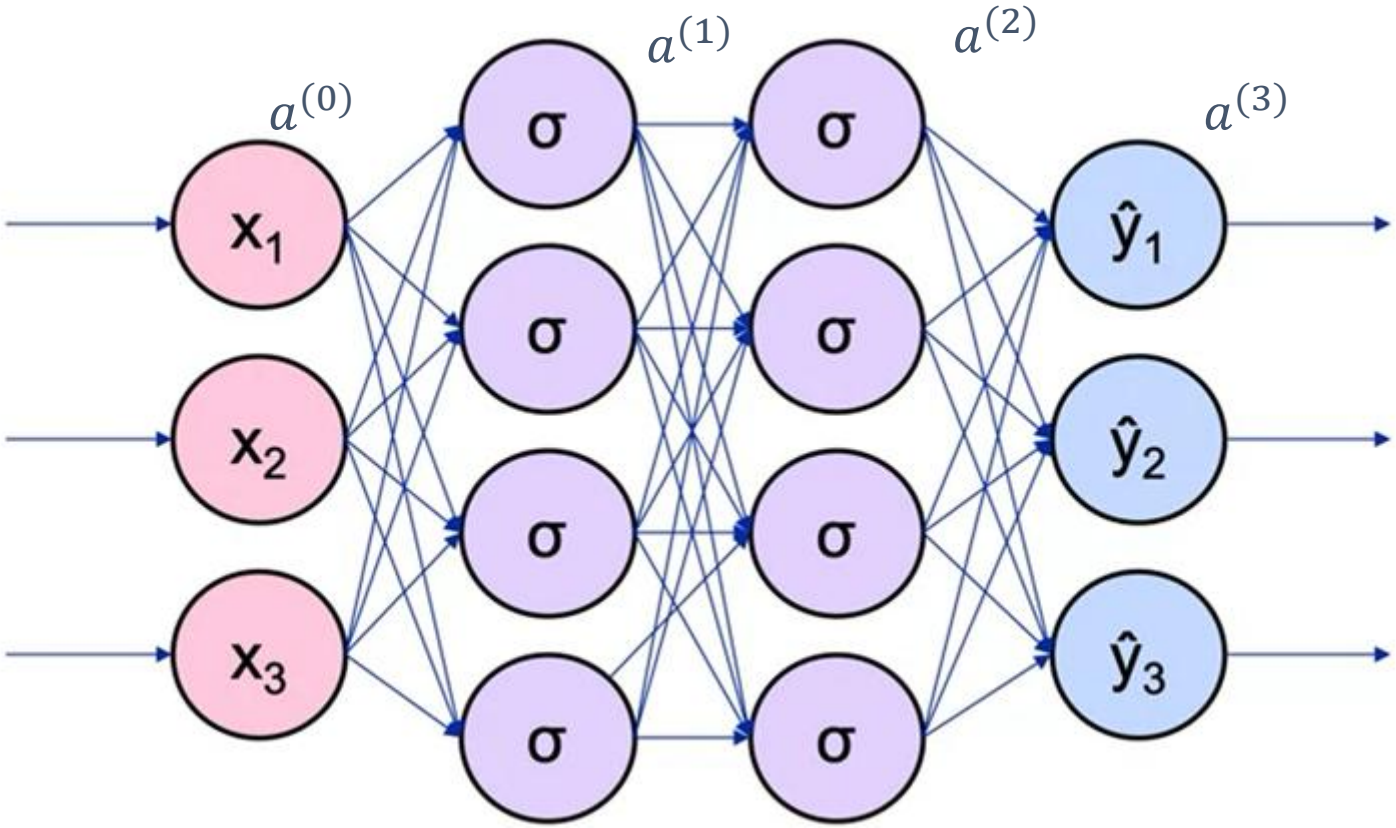
Pesos: Representação Matricial



Net input: Soma ponderada das entradas



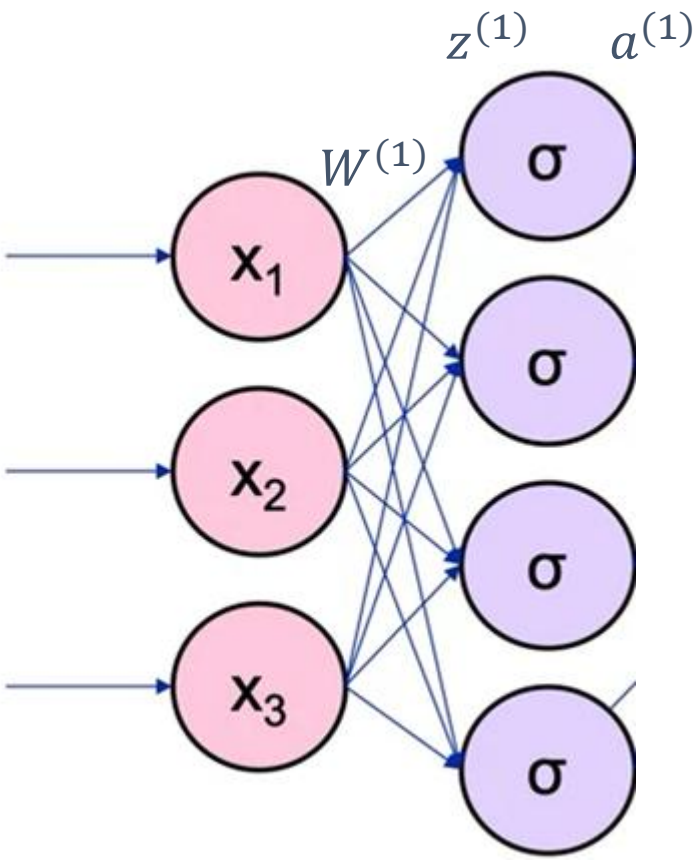
Ativações: Saída par próxima camada



Pesos: Representação Matricial do cálculo

Para um único ponto do conjunto de dados (instância)

- $W^{(1)}$ é uma matriz 3x4
- $z^{(1)}$ é um vetor de 4 elementos
- $a^{(1)}$ é um vetor de 4 elementos
- $x = a^{(0)}$
- $z^{(1)} = x \cdot W^{(1)}$
- $a^{(1)} = \sigma(z^{(1)})$



Continuando o cálculo

- Para um único ponto do conjunto de dados (instância)
- Entrada: vetor x (vetor linha de comprimento 3)
- Saída: vetor $\hat{y}$ (vetor linha de comprimento 3)

$$z^{(1)} = xW^{(1)} \longrightarrow a^{(1)} = \sigma(z^{(1)})$$

$$z^{(2)} = a^{(1)}W^{(2)} \longrightarrow a^{(2)} = \sigma(z^{(2)})$$

$$z^{(3)} = a^{(2)}W^{(3)} \longrightarrow \hat{y} = softmax(z^{(3)})$$

- *softmax*: probabilidade de cada uma das classes